

CONFIDENTIAL



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

FINAL EXAMINATION SEMESTER x 202x/202x

SUBJECT CODE : **SCSR 1013 / SECR 1013**

SUBJECT NAME : **DIGITAL LOGIC**

SECTION :

DURATION : **3 HOURS**

VENUE :

INSTRUCTIONS:

1. There are SIX (6) structure questions.
2. Answer ALL questions in answer booklet.
3. Answer Questions 1(a)(ii), 1(c), 2(a), 2(b), 2(c)(ii), 3(b), and 4(a)(ii), using templates provided in Appendix.

A candidate who is suspected of cheating in examinations is liable to disciplinary action including (but not limited to) suspension or expulsion from the University. All materials and or devices which are found in violation of any examination rules and regulation will be confiscated.

(Please Write Your Lecturer Name and Section In Your Answer Booklet)

Name	
Matric No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **THIRTEEN (13)** printed pages excluding this page.

QUESTION 1 (15 MARKS)

a) Figure 1 shows a logic diagram for a 4-input multiplexer.

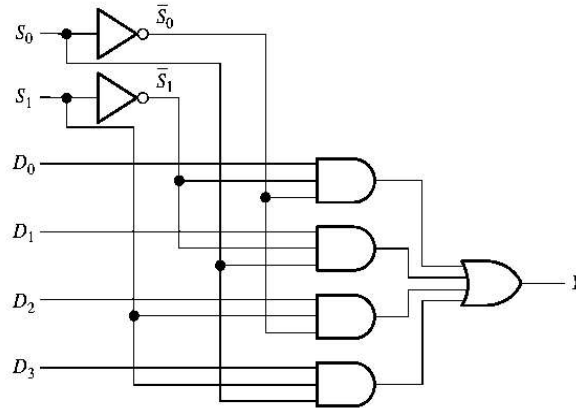


Figure 1

- i. Referring to Figure 1, create the truth table. (3 marks)
- ii. Referring to truth table in a(i), draw the output waveform for the timing diagram in Figure 2. **Answer in Appendix.**

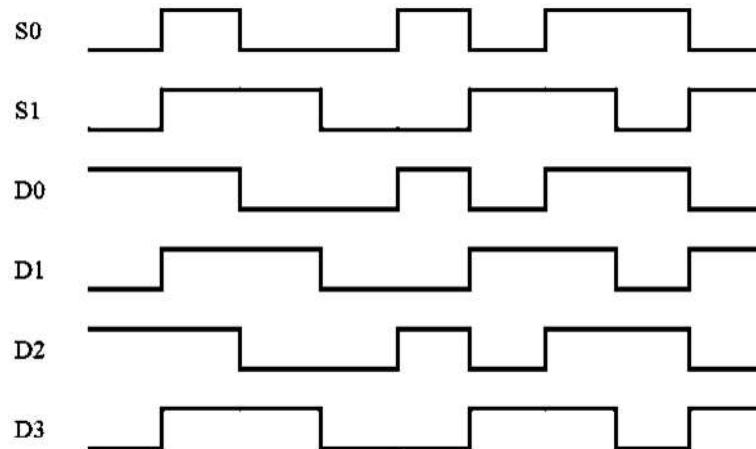


Figure 2

(2 marks)

b) Using parallel adder, the binary numbers 1011 and 0101 are being added.

- i. Draw and label the block diagram for the parallel adder.

(2 marks)

- ii. Calculate the sum of the given binary numbers. Determine the carry in, sum and output carry based on the label defined in b(i).

(3 marks)

c) Identify the statement, either TRUE (T) or FALSE(F). **Answer in Appendix.**

(5 marks)

- i. A full adder is characterized by two inputs and two outputs.
- ii. The modulo-2 sum of four bits can be formed by three exclusive-OR gates connected.
- iii. For the parity checker, when the number of 1s on the inputs is even, the output X is HIGH. When the number of 1s is odd, the output X is LOW.
- iv. The comparator outputs, $A < B$ is HIGH and the other outputs are LOW when A inputs is 0110 and the number on the B inputs is 0011.
- v. Demultiplexer is also known as a data distributor.

QUESTION 2 (16 MARKS)

- a) Draw the output waveform for gated D latch as indicated in Figure 3. **Answer in Appendix.** (2 marks)

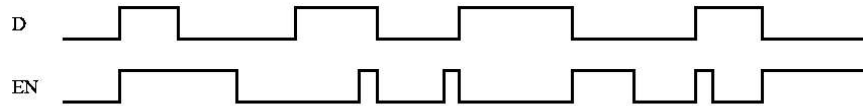


Figure 3

- b) Table 1 shows input combinations for negative clock triggered J-K flip flops. Determine the output Q and its related state for each input. **Answer in Appendix.**

Table 1

Input				
CLK	$\overline{PRE}$	$\overline{CLR}$	J	K
↓	1	0	0	1
↓	1	1	1	1
↓	0	1	1	0
↓	1	1	0	0
↓	1	0	1	1
↓	1	1	0	1
↓	0	1	0	1
↓	1	1	1	0

(4 marks)

- c) Consider a circuit diagram that includes a JK flip-flop (negative clock triggered), a D flip-flop (negative clock triggered), and AND logic gate, as illustrated in Figure 4.

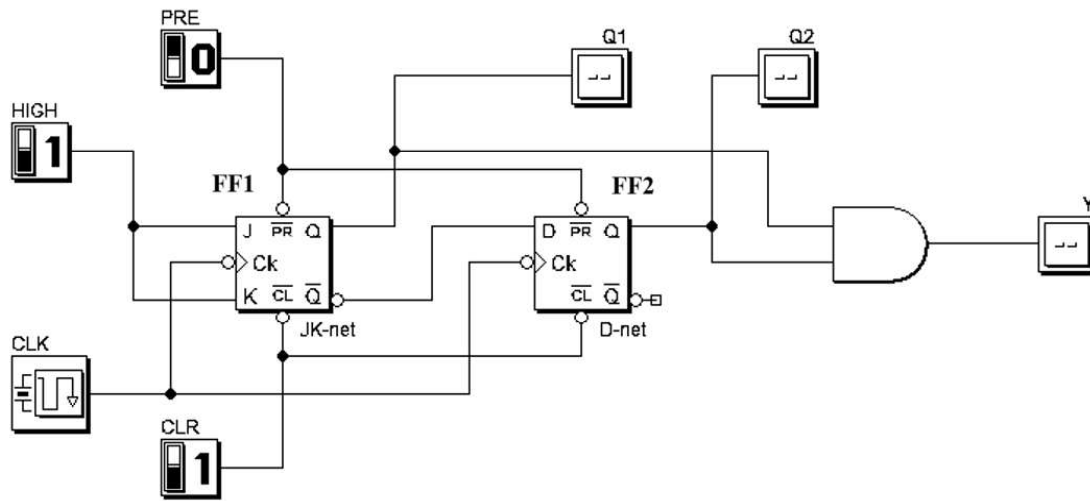


Figure 4: Asynchronous counter

- Write equations for input D and output Y. (2 marks)
- Given the asynchronous and synchronous inputs shown in Table 2, determine the value of input D and output flip-flops Q1, Q2, and output Y. For each output of Q2 identify the correct state. **Answer in Appendix.**

Table 2

CLK	$\overline{PRE}$	$\overline{CLR}$
↓	0	1
↓	1	0
↓	1	1
↓	1	1
↓	0	1
↓	1	1
↓	1	1
↓	1	1

(8 marks)

QUESTION 3 (15 MARKS)

- a) Design and draw a logic circuit for MOD-6 count-up asynchronous counter using negative edge-triggered JK flip-flops with asynchronous inputs.

(9 marks)

- b) Draw the waveform for the counter circuit outputs in a) for 10 clock cycles if $\overline{PRE}$ and $\overline{CLR}$ are HIGH. **Answer in Appendix.**

(6 marks)

QUESTION 4 (14 MARKS)

a) Figure 5 illustrates the timing diagram of a universal bidirectional shift register, where the current input is 3 (in decimal) for a certain period of positive clock edges. The inputs J and K serve as the serial inputs, while D_i represents the parallel data inputs. The parallel outputs of the shift register are denoted as Q_i .

i. List the basic data transfer can be performed by the shift register. (2 marks)

ii. Using Figure 5, complete the timing diagram by drawing the waveforms for parallel outputs. **Answer in Appendix.**

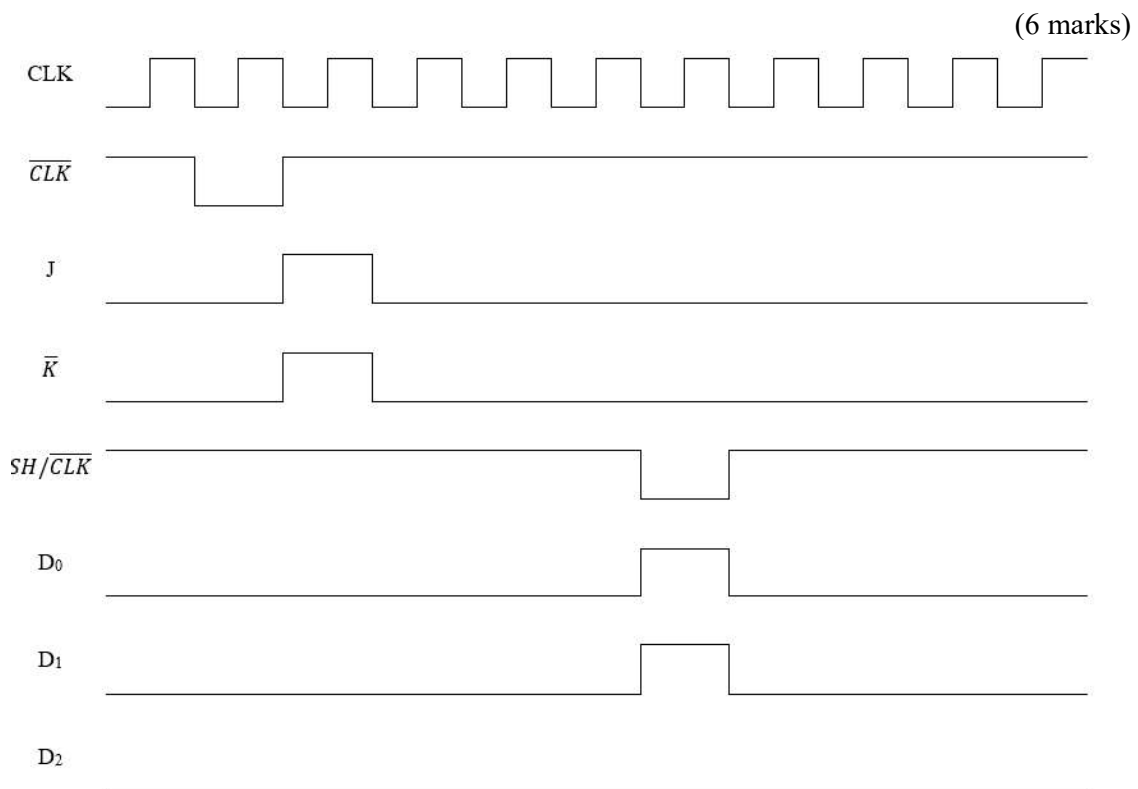


Figure 5

b) Figure 6 shows a Johnson counter.

- i. Construct a complete truth table for the counter with 8 clock cycles. Consider the counter starts with value 4 (decimal).

(4 marks)

- ii. Draw a complete state diagram for this counter.

(2 marks)

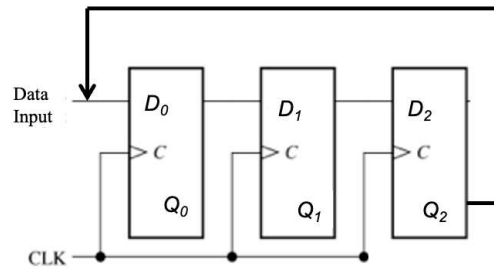


Figure 6: Johnson counter

QUESTION 5 (20 MARKS)

Refer circuit in Figure 7.

- Derive Boolean expressions for the inputs flip-flop J1, K1, J0, K0 and output Y.
(6 marks)
- Produce the next state and flip flop transition table. Use the following Table 3 header.
(8 marks)
- Draw a complete state diagram for the circuit shown.
(6 marks)

Table 3: Next state and flip flop transition table

Input	Present State		Next State		Input Flip Flop				Output
X	Q1	Q0	Q1(+)	Q0(+)	J1	K1	J0	K0	Y

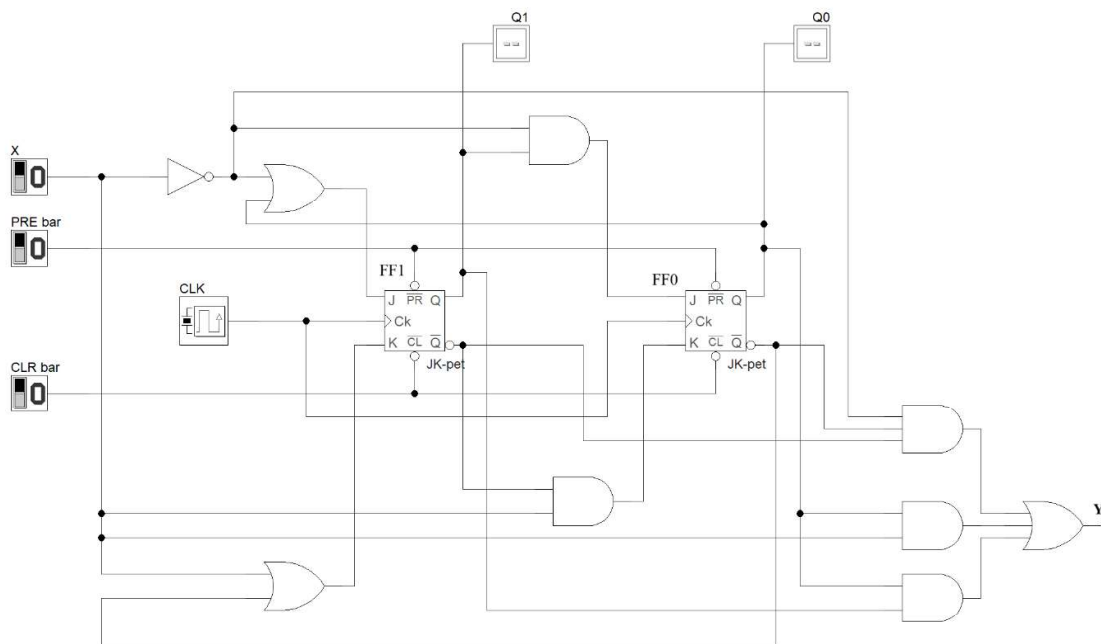


Figure 7: Synchronous counter

QUESTION 6 (20 MARKS)

Refer to state diagram in Figure 8, design a synchronous counter using T flip-flops with input X and output Z.

- a) Construct a complete next state and T flip-flop transition table. Use table header as shown in Table 4.

(8 marks)

- b) Draw K-maps to get minimum Boolean expression using SOP.

(8 marks)

- c) Draw the complete circuit diagram design.

(4 marks)

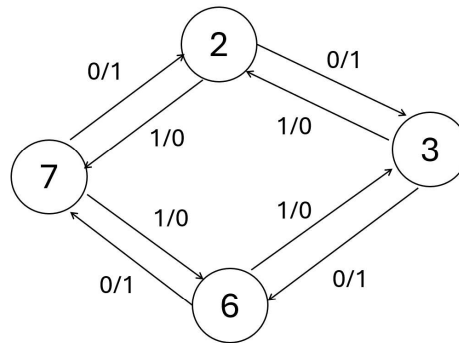


Figure 8

Table 4: Next state and flip flop transition table

Input	Present State			Next State			Input Flip Flop			Output
X	Q2	Q1	Q0	Q2(+)	Q1(+)	Q0(+)	T2	T1	T0	Z

----- END of QUESTIONS -----

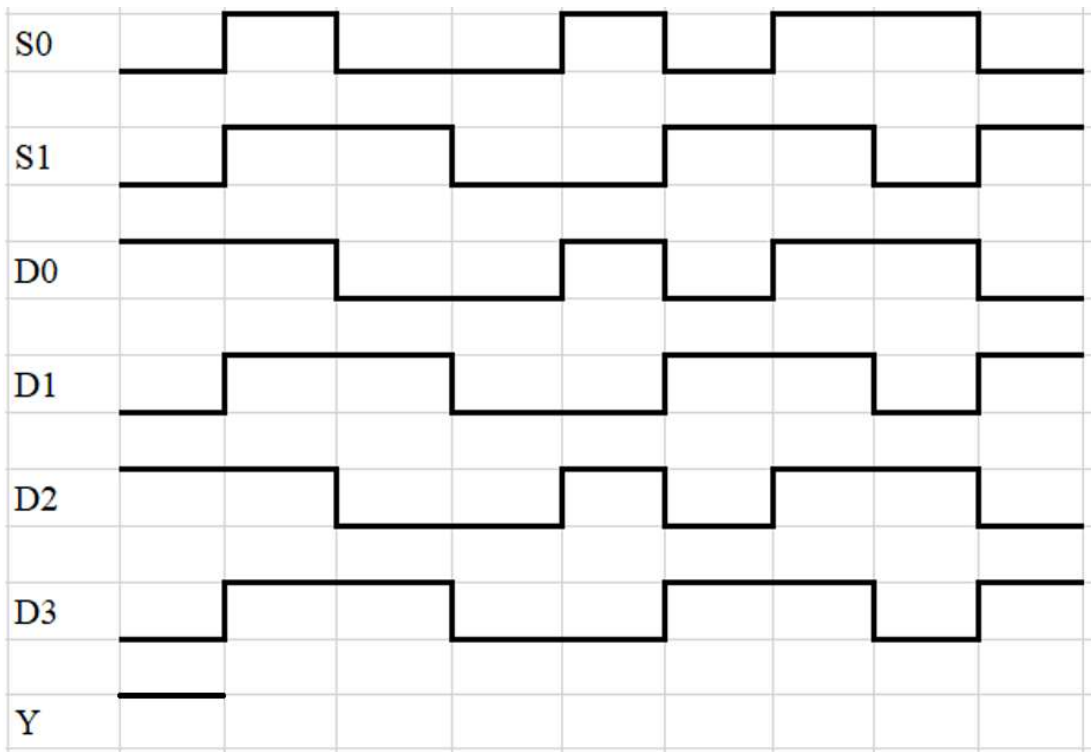
APPENDIX

Name : _____

Matric No : _____

Section : _____

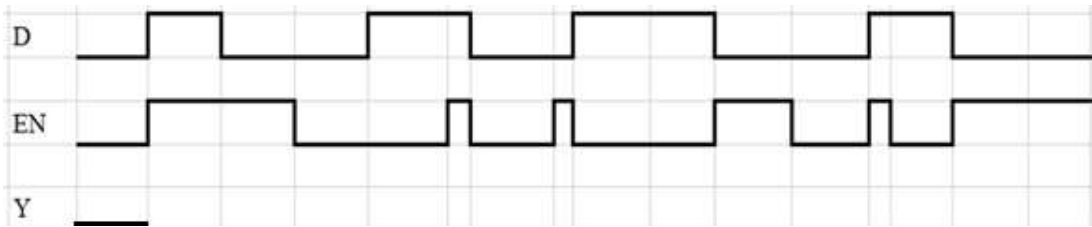
Answer for Question 1(a)(ii)



Answer for Question 1(c)

i.	ii.	iii.	iv.	v.
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Answer for Question 2(a)



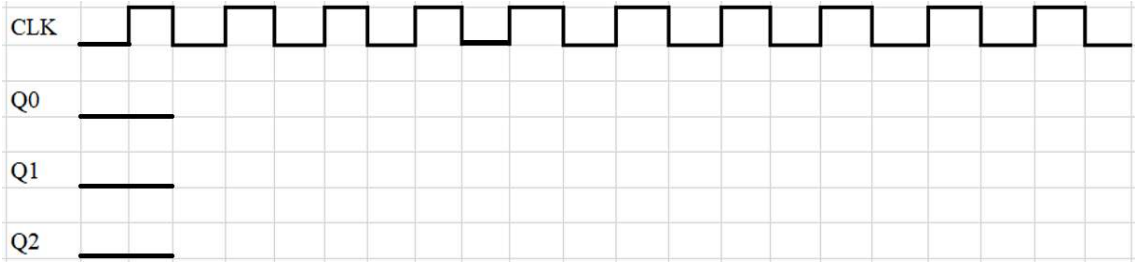
Answer for Question 2(b)

Input					Output	State
CLK	$\overline{PRE}$	$\overline{CLR}$	J	K	Q	Q
↓	1	0	0	1		
↓	1	1	1	1		
↓	0	1	1	0		
↓	1	1	0	0		
↓	1	0	1	1		
↓	1	1	0	1		
↓	0	1	0	1		
↓	1	1	1	0		

Answer for Question 2(c)(ii)

	Input			Output	Output	State	Output
CLK	$\overline{PRE}$	$\overline{CLR}$	D	Q1	Q2	Q2	Y
↓	0	1					
↓	1	0					
↓	1	1					
↓	1	1					
↓	0	1					
↓	1	1					
↓	1	1					
↓	1	1					

Answer for Question 3(b)



Answer for Question 4(a)(ii)

